

GOVINNOVATE

Innovation-to-Procurement Lifecycle Platform

SIH 2026 — Problem Statement SIH26136

Problem Statement:

“Startup friendly public procurement mechanism that enables government departments to identify, pilot, procure, and scale innovative solutions from eligible startups.”

Organization: Government of Maharashtra

Department: Maharashtra State Innovation Society (MSInS), Department of Skills, Employment, Entrepreneurship and Innovation

Category: Software

Theme: Smart Automation

1. PROJECT VISION

GovInnovate is an **Innovation-to-Procurement Lifecycle Platform** that helps government departments discover verified startups, match them with real government problems, technically evaluate their solutions, conduct controlled pilots, validate real-world evidence, collect user feedback, and transition successful innovations into the appropriate procurement pathway.

Core Vision

Government Problem → Verified Innovation → Pilot → Measured Impact → Evidence-Based Procurement → Scale

GovInnovate is **not a replacement for GeM, MahaTenders, or existing government procurement systems.**

Existing procurement systems primarily handle the purchasing/tendering process.

GovInnovate creates the structured **innovation journey before and around procurement:**

Discover → Match → Verify → Evaluate → Pilot → Validate → Feedback → Procure → Scale

2. PROBLEM BEING SOLVED

The current innovation-to-procurement journey can become fragmented across:

- Government problem identification
- Startup discovery
- Startup verification
- Eligibility assessment
- Technical evaluation
- Evidence verification
- Pilot planning
- Pilot monitoring
- Performance validation
- User feedback
- Procurement handoff
- Scale-up decisions

Major Gaps

1. Discovery Gap

Government departments may know their problem but may not easily discover the right startup or innovation.

2. Verification Gap

Startup credentials, claims, certifications, and evidence need structured verification.

3. Evaluation Gap

Technical evaluation can be difficult without domain experts.

4. Pilot Gap

Innovative solutions need controlled PoC/pilot mechanisms before large-scale deployment.

5. Evidence Gap

Government needs measurable evidence instead of relying only on startup claims.

6. Feedback Gap

Pilot success should include actual user/beneficiary experience.

7. Procurement Transition Gap

A successful innovation needs a structured transition from pilot validation into the appropriate procurement pathway.

3. FIVE CORE ACTOR MODULES

GovInnovate has five primary actor groups:

1. Government
2. Startup
3. Admin
4. Expert / Verifier
5. General User / Beneficiary

Responsibility Separation

Actor	Primary Responsibility
Government	Defines problems, manages challenges, monitors pilots and makes authorized procurement decisions
Startup	Provides innovation, submits solutions, provides evidence and executes pilots
Expert	Performs independent technical evaluation and validates evidence/pilot performance
Admin	Governs the platform, manages users, policies, workflows, experts and audit
General User	Provides real-world feedback about pilot/deployed solutions

Core Governance Principle

AI assists. Experts validate. Government decides. Users provide feedback. Admin governs.

4. GOVERNMENT MODULE

Purpose

The Government module allows departments to convert real-world problems into structured innovation challenges, discover suitable startups, manage

evaluations, conduct pilots and move successful solutions toward procurement.

Government Workflow

Admin onboards department



Government officer logs in



Create raw challenge



AI Challenge Structuring



Quality & Policy Validation



Government Review



Publish Challenge



AI Startup Matching



Eligibility Screening



Startup Submissions



Expert Evaluation



PoC Selection



Sandbox Integration



Active Micro-Pilot



Expert Validation



User Feedback



Government Decision



Procurement Handoff



Scale-Up

Government Dashboard

KPIs

- Active Challenges
 - Draft Challenges
 - Startup Submissions
 - Shortlisted Startups
 - Pending Expert Evaluations
 - Active Pilots
 - Completed Pilots
 - Procurement-Ready Solutions
 - Pending Approvals
 - User Feedback
 - Pilot Success Rate
-

Government Navigation

- Dashboard
 - Challenges
 - Startup Submissions
 - AI Matching
 - Expert Evaluations
 - Pilots
 - Evidence & Validation
 - Procurement
 - Milestones & Payments
 - Feedback
 - Reports & Analytics
 - Audit Logs
 - Department Profile
-

5. CHALLENGE CREATION

Government officers should be able to describe a problem naturally.

Example Raw Challenge

“In rural lakes around Nashik, we are facing frequent algal blooms. We currently depend on manual inspection, which is slow and requires field visits. We need a way to detect blooms early and alert the department.”

The system converts this raw description into a structured government challenge.

6. CHALLENGE STRUCTURING ENGINE

Use an LLM such as **Gemini 2.5 Flash** with structured JSON/schema output.

Flow

Raw Government Problem
↓
Challenge Structuring Engine
↓
LLM
↓
Structured Challenge JSON
↓
Validation Engine
↓
Government Review
↓
Publish

Structured Challenge

The system should generate:

- Challenge ID
- Department
- Department Category
- Challenge Title
- Problem Statement
- Current Situation

- Problem Context
- Geographic Scope
- Desired Solution
- Expected Outcome
- Success Definition
- Target Beneficiaries
- Required Capabilities
- Technology Domain
- Integration Requirements
- Infrastructure Requirements
- Data Requirements
- Pilot Location
- Pilot Duration
- Pilot Users
- Pilot Scope
- Pilot Budget
- Success Metrics
- Compliance Requirements
- Evaluation Criteria

Government must be able to edit and approve the AI-generated challenge before publishing.

7. CHALLENGE QUALITY ENGINE

The system checks whether a challenge is sufficiently defined.

Checks

- Problem clarity
- Completeness
- Desired outcome
- Measurable success criteria

- Technical feasibility
- Pilot readiness
- Required capabilities
- Data requirements
- Infrastructure requirements
- Compliance requirements
- Evaluation criteria

The engine should flag missing information rather than silently inventing it.

8. CHALLENGE TAXONOMY ENGINE

Automatically classify challenges by:

- Sector
- Department
- Technology
- Problem type
- Geography
- Keywords
- Required capabilities
- Innovation category

Example

Algal Bloom Detection

- Water Conservation
 - IoT
 - AI
 - Environmental Monitoring
 - Deep Tech
 - Rural Infrastructure
-

9. DUPLICATE CHALLENGE ENGINE

Before publishing a challenge, compare it with existing challenges using semantic similarity.

Example:

New Challenge:

“AI-based lake water quality monitoring.”

Existing Challenge:

“Smart monitoring of rural water bodies.”

System:

Potential Similar Challenge Detected — 87% similarity

Government can review whether to:

- Merge
- Update
- Continue separately

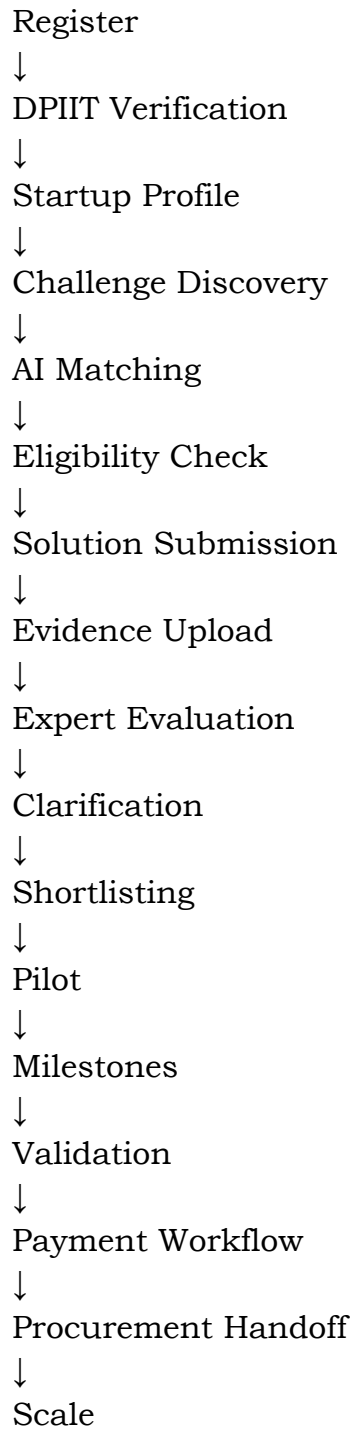
10. STARTUP MODULE

Purpose

The Startup module allows startups to:

- Register
- Verify identity
- Build an innovation profile
- Discover government challenges
- Find AI-matched opportunities
- Check eligibility
- Submit solutions
- Upload evidence
- Respond to expert clarification
- Participate in pilots
- Track milestones
- Receive milestone-based payments through the authorized process
- Build verified performance history
- Progress toward procurement and scale

11. STARTUP FLOW



12. DPIIT VERIFICATION ENGINE

Startup enters:

- DPIIT Recognition Number

- Entity information

The system should use an official validation mechanism when available.

If the prototype uses mock data:

Prototype Verification — Mock Government API

The system should clearly distinguish simulated verification from live government verification.

Result

- Valid
- Invalid
- Unable to Verify

Additional information:

- Recognition number
- Entity
- Recognition date
- Verification timestamp
- Verification source/status

13. STARTUP VERIFICATION ENGINE

Verify:

- DPIIT recognition
- Company identity
- Registration details
- Required registrations
- Startup documentation
- Technical evidence
- Previous performance
- Relevant certifications

Status

- Verified

- Partially Verified
 - Verification Required
 - Invalid / Rejected
-

14. ELIGIBILITY & PROCUREMENT RELAXATION ENGINE

This engine answers:

“For this specific challenge, what eligibility requirements or procurement relaxations may apply?”

It should NOT make blanket claims such as:

“Every DPIIT startup receives every procurement exemption.”

Instead, show:

Requirement	Result
DPIIT Recognition	Verified
Prior Experience	Policy-dependent / may apply
Prior Turnover	Policy-dependent
EMD	Possible exemption subject to applicable rules
Technical Capability Evidence required	
Other Requirements	Challenge-specific

All legal/procurement rules should come from an approved policy knowledge base and remain configurable.

15. STARTUP PROFILE

Startup profile should contain:

Company

- Company name
- Legal entity
- Founder/authorized representative
- Website

- Contact
- Registered location
- Incorporation year

Technology

- Technology domains
- Industry
- Capabilities
- Product
- Innovation
- Deployment readiness

Evidence

- Certifications
- Test reports
- Product documents
- Pilot reports
- Case studies
- Patents/IP
- Performance reports

Past Performance

- Previous pilots
- Outcomes
- Government/enterprise deployments where permitted
- Verified performance history

16. STARTUP MATURITY ENGINE

Classify startup/solution maturity:

**Idea → Prototype → PoC → Validated Product → Pilot Ready →
Deployment Ready → Scale Ready**

This maturity information helps prevent unsuitable startups from being recommended for challenges requiring deployment-ready solutions.

17. AI STARTUP MATCHING ENGINE

Input

Government Challenge

↓

Generate challenge embedding / representation

↓

Compare against startup:

- Technology
- Industry
- Capabilities
- Past work
- Maturity
- Evidence
- Geography
- Eligibility

↓

Rank startups

Matching Score

A prototype score may combine:

- Semantic relevance
- Technology fit
- Sector fit
- Maturity fit
- Past performance
- Geographic suitability
- Eligibility

The exact weights should remain configurable.

Example

Government Challenge:

Algal bloom detection in rural lakes

AI identifies:

AI + IoT + Water Conservation + Environmental Monitoring + Deep Tech

Then ranks suitable startups.

18. AI MATCH EXPLANATION

Do not only show:

Match Score: 92%

Show why:

92% Match

- ✓ IoT sensors
- ✓ Water monitoring experience
- ✓ AI-based anomaly detection
- ✓ Similar previous pilot
- ✓ Required deployment maturity
- ✓ Eligibility verified

This makes the AI recommendation explainable.

19. AI SUBMISSION ASSISTANT

Startup enters a raw solution description.

AI structures it into:

- Solution Overview
- Problem Addressed
- Technical Approach
- Innovation
- Implementation Plan
- Pilot Plan

- Expected Outcomes
- Evidence

The startup must review and approve the final submission.

Critical Rule

AI must **not fabricate**:

- Certifications
 - Performance metrics
 - Customers
 - Government deployments
 - Patents
 - Experience
 - Compliance status
-

20. EVIDENCE VAULT

Startup uploads:

- Certificates
- Test reports
- Product specifications
- Demo videos
- Pilot reports
- Accuracy reports
- Case studies
- Patents/IP
- Compliance documents

Evidence Status

- Pending Review
- Verified
- Verified with Conditions
- Needs Clarification

- Rejected / Insufficient
-

21. EVIDENCE EXTRACTION ENGINE

Documents are processed using:

Document → **Parsing** → **Chunking** → **Embeddings** → **PostgreSQL/pgvector** → **Retrieval** → **LLM**

Extract:

- Claimed accuracy
- Metrics
- Certifications
- Technology
- Locations
- Pilot duration
- Performance
- Deployment information

Example

Startup claims:

90% accuracy

Evidence reports:

87.4%

System generates:

Claim–Evidence Discrepancy Detected

and routes it to the expert.

22. EXPERT / VERIFIER MODULE

Purpose

Experts provide independent technical validation.

The expert answers:

“Is this innovation technically credible, and does the available evidence support its claims?”

Experts should not make procurement decisions independently.

23. EXPERT PROFILE

- Expert ID
 - Name
 - Organization
 - Domain
 - Specialization
 - Years of Experience
 - Qualifications
 - Certifications
 - Previous evaluation history
 - Active assignments
 - Conflict-of-interest status
-

24. EXPERT DASHBOARD

KPIs:

- Pending Reviews
- Assigned Evaluations
- Active Evaluations
- Completed Evaluations
- Pilot Validations
- Clarification Requests

Review Queue

Each assignment displays:

- Startup
- Challenge

- Category
 - DPIIT status
 - Pilot stage
 - Assignment date
 - Deadline
 - Status
-

25. EXPERT ASSIGNMENT ENGINE

The platform can recommend experts based on:

- Domain
- Technology
- Experience
- Sector
- Previous evaluations
- Workload
- Availability
- Conflict-of-interest status

Admin/program authority can assign the final expert.

26. CONFLICT-OF-INTEREST ENGINE

Before evaluation, expert must declare:

“I declare that I have no conflict of interest with this startup or evaluation.”

Store:

- Declaration
- Timestamp
- Expert
- Evaluation
- Status

This becomes part of the audit trail.

27. EXPERT EVALUATION

Prototype evaluation criteria:

Criterion	Weight
Technical Feasibility	25%
Innovation / Novelty	20%
Deployment Readiness	20%
Scalability	15%
Team Capability	10%
Evidence / Validation Quality	10%

Important: These are prototype/configurable evaluation weights, not an official Maharashtra evaluation standard.

28. EXPERT EVIDENCE REVIEW

Expert can review:

- Technical reports
- Certifications
- Test results
- Product documentation
- Demo videos
- Pilot reports
- Accuracy reports
- Case studies
- Compliance evidence

Expert can mark:

- ✓ Verified
- ✓ Verified with Conditions
- △ Needs Clarification
- ✗ Insufficient Evidence

29. EXPERT CLARIFICATION

If evidence is insufficient:

Expert



Clarification Request



Startup Notification



Startup Uploads Evidence



Expert Review



Status Updated

The startup cannot edit the expert's score.

30. EXPERT RECOMMENDATION

Possible outcomes:

- Recommend
- Recommend with Conditions
- Needs Improvement
- Do Not Recommend
- Recommend for Scale

The expert provides:

- Score
 - Strengths
 - Weaknesses
 - Evidence findings
 - Technical concerns
 - Conditions
 - Recommendation
-

31. PILOT VALIDATION

Expert validates pilot metrics.

Example:

Detection Accuracy

Target: 90%

Actual: 87.4%

Response Time

Target: <10 minutes

Actual: 7 minutes

Sensor Uptime

Target: ≥95%

Actual: 97.2%

The system should preserve:

Target → Actual → Evidence → Expert Validation → Result

32. EXPERT PERMISSIONS

Expert CAN:

- ✓ Evaluate
- ✓ Verify evidence
- ✓ Request clarification
- ✓ Validate pilot metrics
- ✓ Add comments
- ✓ Submit recommendations

Expert CANNOT:

- ✗ Award procurement
 - ✗ Release payment
 - ✗ Modify procurement law
 - ✗ Edit another expert's evaluation
 - ✗ Independently select the final supplier
-

33. ADMIN MODULE

Purpose

Admin governs the platform rather than acting as the technical evaluator or procurement authority.

There should be two conceptual levels.

Platform Super Admin

Recommended for MSInS / authorized program authority.

Department Admin / Nodal Officer

Manages a specific department.

34. PLATFORM SUPER ADMIN

Responsibilities:

- Government department onboarding
- Department approval
- User and role management
- Expert pool management
- Platform workflow configuration
- Evaluation template configuration
- Policy knowledge base
- Document templates
- Risk configuration
- SLA configuration
- Notifications
- System analytics
- Audit monitoring
- AI monitoring
- Escalations
- Dispute management
- Security/RBAC

35. DEPARTMENT ADMIN

Responsibilities:

- Manage department users
- Create/manage challenges
- Review AI-generated challenge structures
- Publish challenges
- View submissions
- Request expert evaluation
- Monitor pilots
- Review validation
- Initiate procurement handoff
- View department analytics

36. ADMIN ENGINES

1. Department Onboarding Engine
 2. User & RBAC Engine
 3. Expert Pool Management Engine
 4. Expert Assignment Engine
 5. Challenge Moderation Engine
 6. Evaluation Template Engine
 7. Policy & Rule Configuration Engine
 8. Document/Template Management Engine
 9. Risk Configuration Engine
 10. SLA & Notification Engine
 11. Audit & Compliance Engine
 12. Analytics Engine
 13. AI Monitoring Engine
 14. Knowledge Base/RAG Governance Engine
 15. Dispute & Escalation Engine
-

37. ADMIN MUST NOT

Admin should not:

- Independently technically evaluate startups
- Change submitted expert scores
- Award procurement independently
- Bypass procurement rules
- Modify government law/policy casually
- Release payments outside the authorized workflow

Admin = **Governance + Configuration + Oversight**

38. GENERAL USER / BENEFICIARY MODULE

Purpose

The General User module provides a real-world feedback layer.

Users can include:

- Citizens
- Beneficiaries
- Service users
- Employees
- Other permitted pilot users

They provide feedback on the actual experience of an innovation.

39. GENERAL USER FLOW

Access Pilot/Solution



View Service/Solution



Provide Feedback



Rate Experience



Report Issue / Suggestion



AI Feedback Analysis



Government/Admin Review



Improvement Action

40. FEEDBACK TYPES

Rating

1–5 rating.

Structured Feedback

Examples:

- Was it easy to use?
- Was the service reliable?
- Did it solve the intended problem?
- Was the response time acceptable?

Open Feedback

“What problems did you face?”

Issue Reporting

- Technical problem
 - Service unavailable
 - Incorrect result
 - Accessibility problem
 - Slow response
 - Other
-

41. FEEDBACK ANALYSIS ENGINE

NLP categorizes feedback into:

Positive

“Very useful service.”

Negative

“System frequently failed.”

Suggestion

“Add mobile notifications.”

Issue

“Sensor data was delayed.”

The system can identify recurring patterns.

42. FEEDBACK DASHBOARD

Government/Admin can view:

- Total Feedback
- Average Rating
- Satisfaction %
- Reported Issues
- Common Complaints
- Suggested Improvements
- Feedback by Pilot
- Feedback Trends

Feedback is an **impact signal**, not an automatic procurement decision.

43. COMPLETE PILOT LIFECYCLE

The pilot state machine should be:

Challenge

↓

PoC Selection

↓

Sandbox Integration

↓

Active Micro-Pilot

↓

Expert Validation



User Feedback



Government Approval



Procurement Handoff



Scale-Up

44. PILOT WORKSPACE

Each pilot contains:

- Pilot ID
- Challenge
- Startup
- Government department
- Expert
- Start date
- End date
- Location
- Scope
- Budget
- KPIs
- Milestones
- Evidence
- Issues
- Expert validation
- User feedback
- Government decision

45. MILESTONE ENGINE

Example:

M1

System Installation

M2

Sandbox Integration

M3

Field Deployment

M4

Performance Validation

M5

Final Pilot Report

Each milestone contains:

- Target
- Evidence
- Due date
- Approval status
- Payment status

46. PAYMENT GOVERNANCE

Use the terminology:

Milestone-Based Payment Governance

rather than claiming a real financial escrow system unless an actual escrow mechanism is implemented.

Flow

Startup submits milestone evidence

↓

Department/authorized evaluator reviews

↓

Milestone approved



Payment workflow triggered



Payment status recorded

Possible statuses:

- Pending
- Evidence Submitted
- Under Review
- Approved
- Payment Processing
- Paid
- Rejected
- Revision Required

47. PROCUREMENT HANDOFF

After successful pilot:

Pilot Evidence



Expert Validation



User Feedback



Government Review



Procurement Route Assessment



Applicable Government Procurement Channel



Procurement Process



Scale

GovInnovate does not replace:

- GeM
- MahaTenders

- Department procurement systems

It creates the structured innovation pathway and hands successful solutions into the appropriate procurement mechanism.

48. RAG / KNOWLEDGE ENGINE

RAG should answer:

“What do the approved documents and evidence say?”

RAG should NOT replace the database or rule engine.

Knowledge Base

Possible documents:

- Maharashtra Procurement Manual
 - Maharashtra Government Resolutions
 - Startup policy documents
 - Startup documentation
 - Pilot reports
 - Expert evaluations
 - Procurement guidelines
 - Standard templates
 - NDA/IP documents
 - Cybersecurity requirements
 - Risk-management documents
-

49. RAG ARCHITECTURE

Documents



Document Parser



Chunking



Embeddings



PostgreSQL + pgvector



LlamaIndex Retrieval



Relevant Evidence



LLM



Answer + Sources

Critical Rule

If sufficient evidence does not exist:

“I could not find sufficient evidence in the approved knowledge base.”

The system must not invent government rules.

50. DIFFERENCE BETWEEN AI COMPONENTS

RAG

Answers:

“What does the policy/evidence document say?”

Embedding/NLP Matching

Answers:

“Which startup is relevant to this problem?”

Rule Engine

Answers:

“Given these conditions, which configured eligibility/procurement rule may apply?”

Database

Answers:

“What is the current state of this startup/challenge/pilot/payment?”

These are separate functions.

51. AI ENGINES IN GOVINNOVATE

Core AI/automation engines:

1. Challenge Structuring Engine
 2. Challenge Quality Engine
 3. Challenge Taxonomy Engine
 4. Duplicate Challenge Engine
 5. Startup Matching Engine
 6. Eligibility Engine
 7. Procurement Relaxation Engine
 8. AI Submission Assistant
 9. Evidence Extraction Engine
 10. Evidence Retrieval/RAG Engine
 11. Expert Assignment Recommendation Engine
 12. Risk Engine
 13. Feedback Analysis Engine
 14. Policy Assistant
 15. AI Monitoring Engine
-

52. RISK ENGINE

The platform can calculate a configurable pilot risk score using factors such as:

- Startup maturity
- Technology readiness
- Pilot complexity
- Data sensitivity
- Deployment environment

- Dependency on infrastructure
- Safety/compliance requirements
- Evidence quality

Output

Low / Medium / High Risk

A **Micro-Pilot Risk Matrix** should explain why the risk score was generated.

All weights should be labelled:

Configurable Prototype Methodology

They should not be represented as official government risk standards unless officially sourced.

53. DOCUMENT & TEMPLATE ENGINE

Provide standardized templates for:

- Mutual NDA
- IP protection
- Pilot agreement
- Data-sharing clauses
- Cybersecurity requirements
- Risk-management checklist
- Evaluation form
- Pilot report
- Validation report
- Procurement handoff report

Templates should be version controlled.

54. AUDIT ENGINE

Every critical action should generate an audit record.

Examples:

- User login

- Startup verification
- Challenge creation
- Challenge modification
- Challenge publication
- AI recommendation
- Expert assignment
- Conflict declaration
- Evaluation submission
- Evidence verification
- Clarification request
- Pilot creation
- Milestone approval
- Payment status change
- Government decision
- Procurement handoff

Audit log:

Who → What → When → Object → Previous State → New State

55. ROLE-BASED ACCESS CONTROL

Government

Can access:

- Own department challenges
- Startup submissions
- Evaluations
- Pilots
- Validation
- Feedback
- Procurement handoff

Startup

Can access:

- Own profile
- Own submissions
- Own evidence
- Own evaluations/status
- Own pilots
- Own milestones

Expert

Can access:

- Assigned evaluations
- Relevant startup evidence
- Assigned pilots
- Evaluation forms

Admin

Can access:

- Platform governance
- Configuration
- Cross-platform analytics
- Audit
- Users
- Experts
- Departments

General User

Can access:

- Public/eligible pilot information
- Feedback forms
- Own feedback status where applicable

56. SECURITY PRINCIPLES

Implement:

- RBAC
 - Authentication
 - Authorization
 - Secure file storage
 - Encryption where appropriate
 - Audit logs
 - Input validation
 - API authentication
 - Rate limiting
 - Data minimization
 - Privacy controls
 - Role-based document access
 - Separation of confidential procurement data
-

57. CORE DATA MODEL

Startup

- startup_id
- company_name
- DPIIT_number
- legal_entity
- incorporation_date
- technology_tags
- industry
- maturity_level
- verification_status
- evidence

Government Challenge

- challenge_id

- department_id
- title
- problem_statement
- desired_solution
- technology_domain
- geography
- pilot_budget
- pilot_duration
- success_metrics
- eligibility_requirements
- status

Submission

- submission_id
- startup_id
- challenge_id
- proposal
- technical_approach
- pilot_plan
- evidence
- status

Expert

- expert_id
- specialization
- qualifications
- experience
- availability
- COI_status

Evaluation

- evaluation_id

- startup_id
- challenge_id
- expert_id
- criteria_scores
- evidence_findings
- comments
- recommendation
- timestamp

Pilot

- pilot_id
- challenge_id
- startup_id
- expert_id
- department_id
- location
- start_date
- end_date
- KPIs
- milestones
- status

Milestone

- milestone_id
- pilot_id
- description
- target
- evidence
- approval_status
- payment_status

Feedback

- feedback_id
- pilot_id
- user_type
- rating
- category
- comment
- sentiment
- issue_type
- timestamp

Audit Log

- audit_id
- actor_id
- actor_role
- action
- object_type
- object_id
- timestamp
- previous_state
- new_state

58. TECHNOLOGY ARCHITECTURE

Recommended prototype stack:

Frontend

- React / Next.js
- TypeScript
- Tailwind CSS
- Component library

Backend

- Node.js

- Express.js

Database

- PostgreSQL
- pgvector

AI

- LlamaIndex
- Gemini 2.5 Flash
- Embeddings
- NLP / semantic matching

AI Service

For a stronger architecture:

React / Next.js



Node.js + Express



PostgreSQL + pgvector



Python AI Service



LlamaIndex

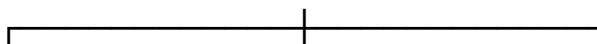


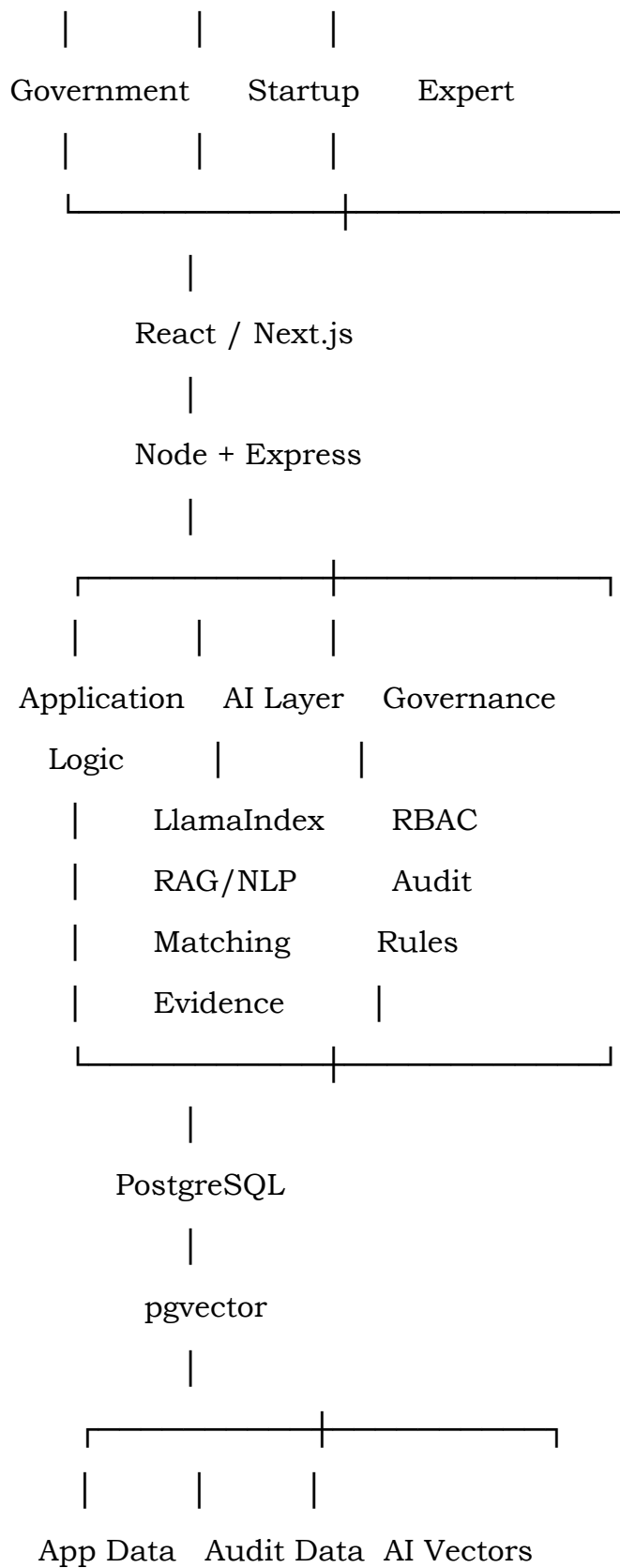
Gemini / Embedding Models

Alternatively, LlamaIndex.TS can be used if keeping the AI stack fully JavaScript/TypeScript.

59. COMPLETE TECHNICAL FLOW

USERS





60. EXTERNAL INTEGRATIONS

Potential integrations:

Startup Verification

- DPIIT / Startup India
- MSInS

For prototype:

Mock API / simulated integration

unless a genuine API is available and authorized.

Procurement

- GeM
- Maharashtra e-Procurement / MahaTenders
- Other authorized procurement channels

GovInnovate should primarily perform **handoff/integration**, not duplicate the procurement platform.

Other

- Email
- SMS/notifications
- PDF generation
- Maps/location services
- Secure document storage

61. END-TO-END EXAMPLE

Challenge

Government Department:

“Detect algal blooms in rural lakes before they become severe.”

Step 1 — Challenge Creation

Government enters natural-language problem.

Step 2 — AI Structuring

System generates:

AI + IoT + Water Conservation + Environmental Monitoring

Step 3 — Startup Matching

System finds suitable startups.

Example:

AquaSense Technologies — 92% Match

Reason:

- IoT sensors
- Water monitoring
- AI anomaly detection
- Relevant previous pilot
- Required maturity

Step 4 — Verification

DPIIT status:

Verified

Evidence:

Partially Verified

Step 5 — Expert Evaluation

Expert evaluates:

Technical Feasibility → 88

Innovation → 91

Deployment Readiness → 76

Scalability → 84

Team Capability → 82

Evidence Quality → 79

Weighted score:

83.7 / 100

Recommendation:

Recommend with Conditions

Step 6 — Pilot

Pilot stages:

PoC

→ Sandbox

→ Micro-Pilot

Step 7 — Validation

Detection accuracy:

Target: 90%

Actual: 87.4%

Response time:

Target: <10 min

Actual: 7 min

Sensor uptime:

Target: ≥95%

Actual: 97.2%

Step 8 — User Feedback

Users report:

- Easy to use
- Alerts useful
- Occasional sensor delay

AI identifies:

Positive overall sentiment + recurring sensor-delay issue

Step 9 — Government Decision

Government reviews:

- Expert score
- Pilot metrics
- Evidence
- User feedback
- Risk
- Cost

Step 10 — Procurement Handoff

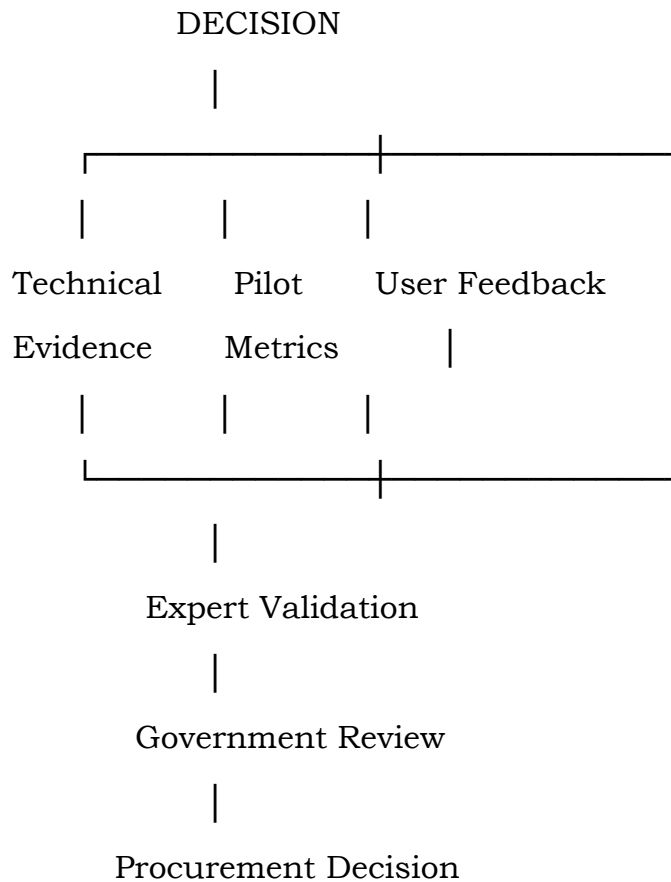
Successful solution moves into the applicable procurement pathway.

Step 11 — Scale

Solution expands to additional locations.

62. CORE DECISION MODEL

GovInnovate should combine five evidence layers:



AI should assist this process but should not independently make the procurement decision.

63. THE INNOVATION OF GOVINNOVATE

The platform is not simply:

- A startup directory
- A tender portal
- A marketplace
- A chatbot
- A generic AI dashboard
- A GeM replacement

Its core innovation is the **structured innovation-to-procurement lifecycle**.

Key Innovations

1. Problem-to-Startup Matching

Government problem → AI → relevant startups.

2. Verified Innovation Profile

Startup verification and evidence follow the startup throughout the lifecycle.

3. Expert-in-the-Loop Validation

AI recommendations are technically reviewed by domain experts.

4. Pilot-First Procurement Path

Innovative solutions can be evaluated through controlled pilots before scaling.

5. Evidence-Based Decision Making

Claims → Evidence → Expert Validation → Pilot Metrics.

6. User Feedback Loop

Real-world users contribute evidence about service quality and impact.

7. Transparent Lifecycle

Every critical decision can be audited.

8. Procurement Handoff

Successful innovation moves from GovInnovate into the applicable procurement channel.

64. CORE LIFECYCLE

FULL VERSION

**DISCOVER → MATCH → VERIFY → EVALUATE → PILOT → VALIDATE →
FEEDBACK → PROCURE → SCALE**

SHORT VERSION

Problem → Innovation → Evidence → Impact → Procurement

65. CORE AI + HUMAN MODEL

GOVERNMENT PROBLEM



AI DISCOVERY



AI MATCHING



VERIFICATION



EXPERT EVALUATION



PILOT



EXPERT VALIDATION



USER FEEDBACK



GOVERNMENT DECISION



PROCUREMENT



SCALE



66. CORE GOVERNANCE PRINCIPLES

Principle 1

AI should assist, not autonomously award procurement.

Principle 2

Government remains the decision authority.

Principle 3

Experts provide independent technical validation.

Principle 4

Startup claims should be evidence-backed.

Principle 5

Procurement rules should come from approved policy sources.

Principle 6

Prototype assumptions must be clearly labelled as configurable.

Principle 7

Every important action should be auditable.

Principle 8

GovInnovate complements existing procurement infrastructure rather than replacing it.

67. SUCCESS METRICS FOR THE PLATFORM

Possible platform-level metrics:

Discovery

- Time to discover relevant startups
- Number of matched startups
- Match relevance

Verification

- Verification completion time
- Evidence verification rate
- Invalid/insufficient claims detected

Evaluation

- Evaluation turnaround time
- Expert participation
- Clarification rate

Pilot

- Pilot completion rate
- KPI achievement
- Pilot failure rate
- Time from selection to pilot

Impact

- User satisfaction
- Issues resolved
- Measured improvement

Procurement

- Pilot-to-procurement conversion
- Time from validation to procurement handoff
- Number of innovations scaled

These are proposed platform KPIs, not official government benchmarks.

68. WHAT MAKES THE SYSTEM TRUSTWORTHY

GovInnovate should provide:

Explainability

→ Why was this startup matched?

Evidence

→ What supports the startup's claim?

Human Validation

→ Which expert verified it?

Traceability

→ Who made each decision?

Pilot Evidence

→ Did it actually work?

User Feedback

→ Did people benefit?

Governance

→ Was the appropriate process followed?

This transforms:

“Startup says it works.”

into:

“Startup claims → Evidence → Expert validation → Pilot result → User feedback → Government decision.”

69. ONE-LINE PITCH

GovInnovate is an AI-assisted innovation-to-procurement platform that connects government problems with verified startups, validates solutions through experts and controlled pilots, captures real-world feedback, and enables evidence-based transition into procurement and scale.

70. FINAL SYSTEM STORY**Government**

I have a problem.

↓

GovInnovate AI

I can find relevant innovations.

↓

Startup

I have a solution and evidence.

↓

Verification

The startup and evidence are checked.

↓

Expert

The solution is technically evaluated.

↓

Pilot

Let's test it in the real world.

↓

User

Here is my actual experience.

↓

Validation

Here is the measured performance.

↓

Government

The evidence supports the decision.

↓

Procurement

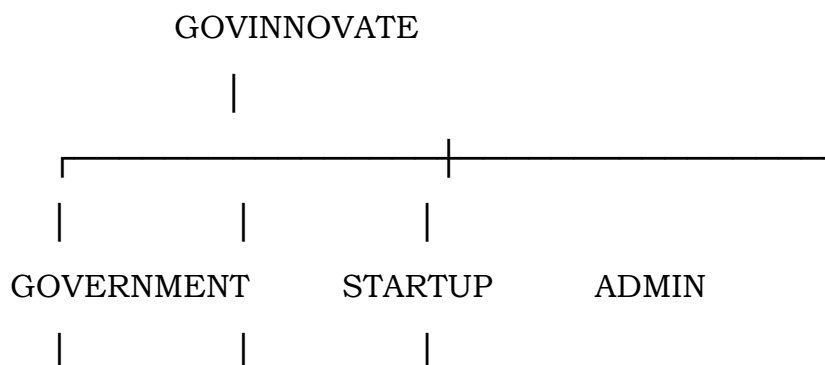
Move into the applicable procurement pathway.

↓

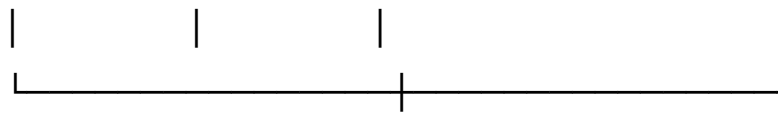
Scale

Deploy the proven innovation more widely.

71. FINAL GOVINNOVATE MODEL



Creates Problems Provides Ideas Governs Platform



AI DISCOVERY



AI MATCHING



VERIFICATION



EXPERT EVALUATION



PILOT



EXPERT VALIDATION



USER FEEDBACK



EVIDENCE ANALYSIS



GOVERNMENT DECISION



PROCUREMENT HANDOFF



SCALE



REAL-WORLD IMPACT

GOVINNOVATE

From Government Problem to Proven Innovation.

From Proven Innovation to Public Impact.

From Public Impact to Scalable Procurement.